Give a Graph Brain to Your LLMs

-- Making LLM Smarter with Knowledge Graphs

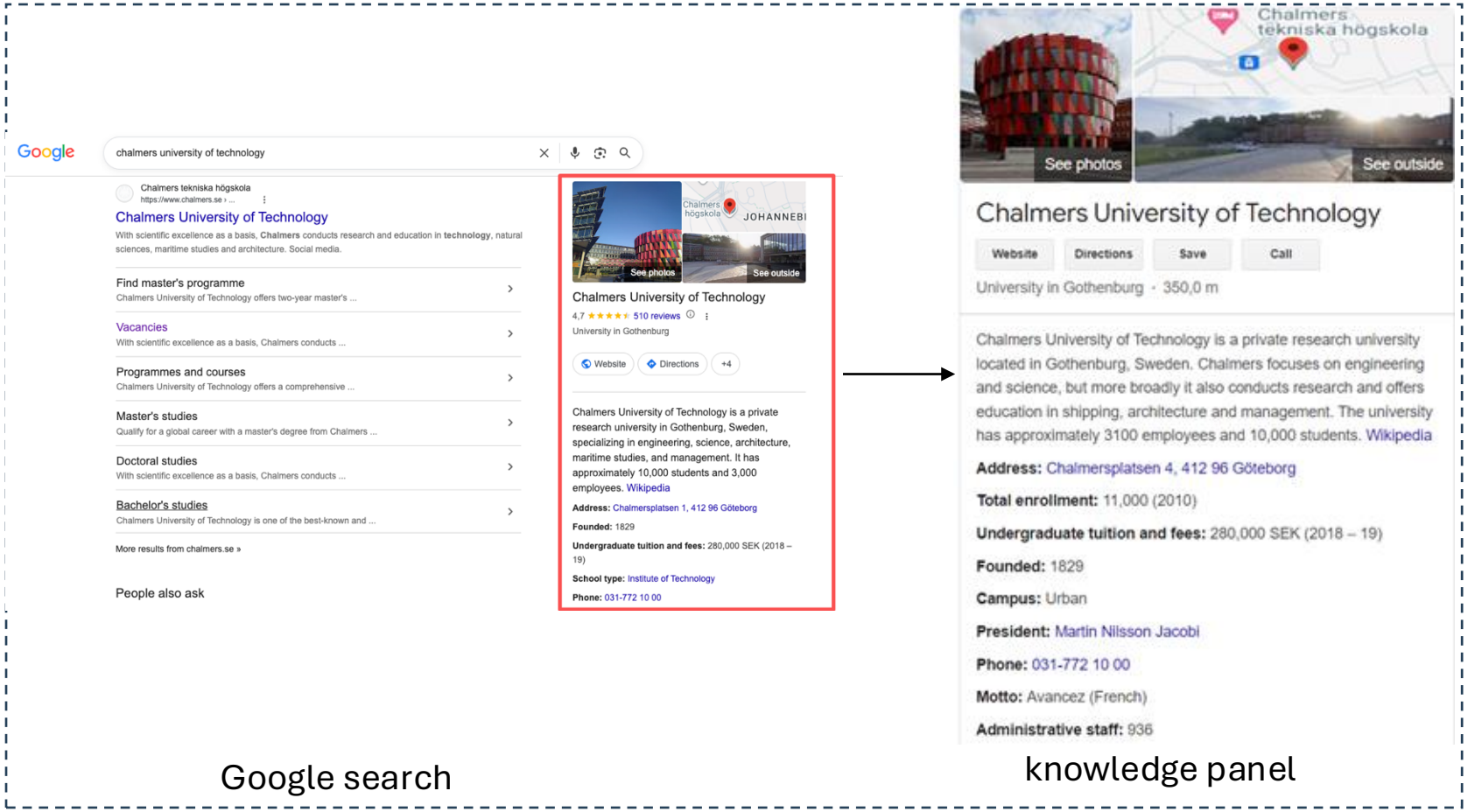
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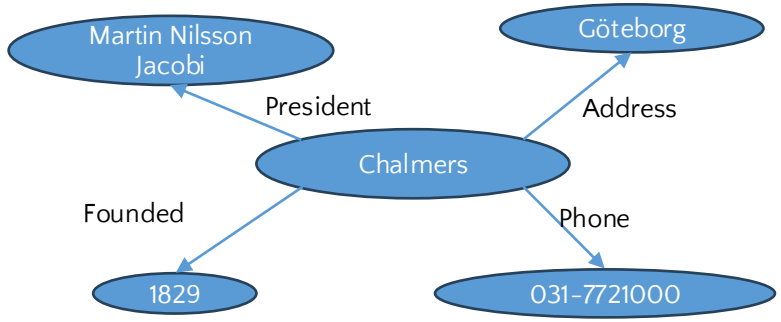
Shuai Wang

Background

Knowledge Graph

A **Knowledge Graph (KG)** is a structured representation of knowledge.





knowledge graph

Connectivity: A Key Strength of Knowledge Graphs

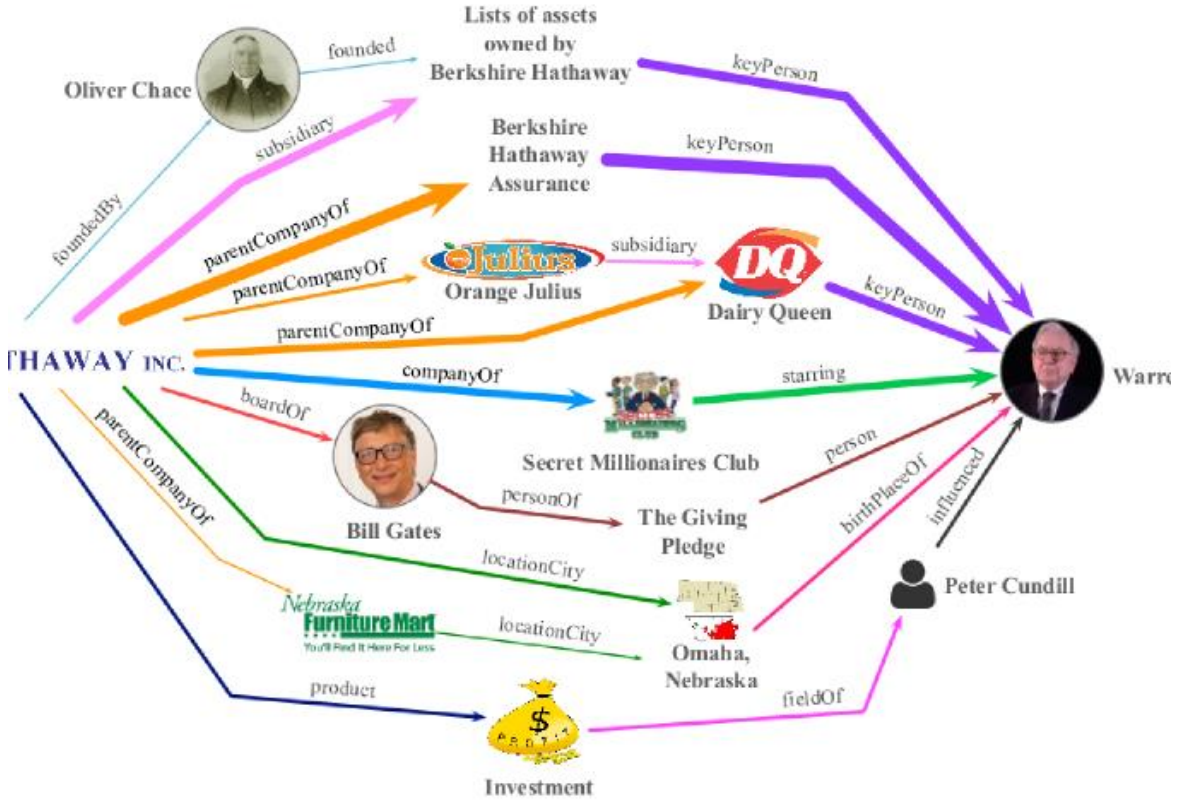
❖ KG connects entities through relationships, forming a network.

- For example:

Person → Company → Location → Product → Organization

➢ We can follow the links to:

- Understand complex relationships
- Connect information from different domains
- Discover indirect, multi-hop connections
- Reveal hidden relationships



❖ Connectivity turns isolated facts into a network that can support discovery and reasoning.

Knowledge Graph in Search

-- Connecting Meaning, Not Just Words

➢ The Problem: Keyword Matching Is Not Enough

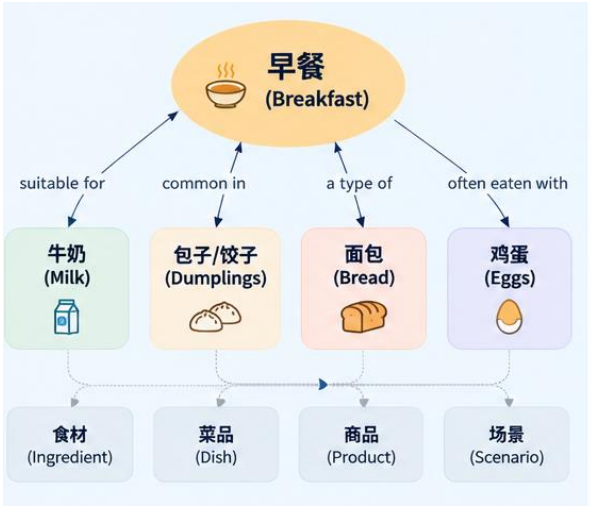
When a user searches for **“breakfast”**, many relevant products may **not contain the word “breakfast” in their names**.

- For example:

  “Breakfast” ≠ Dumplings ≠ Milk ≠ Bread ≠ Eggs



➢ How Does the Knowledge Graph Help?



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https://tech.meituan.com/2018/11/22/meituan-brain-nlp-01.html

A Real-World Knowledge Graph Use Case

https://digitalinsights.qiagen.com/wp-content/uploads/2026/04/Drug-Discover-with-BKB-Biopharma-Application-Note-0525-QPRO-10561.pdf?cmpid=CM_QDI_DS_QIAGEN-Discovery-Platform_0526_SEA_Google_Global

https://digitalinsights.qiagen.com/qiagen-discovery-platform/?cmpid=CM_QDI_DS_QIAGEN-Discovery-Platform_0526_SEA_Google_Global&utm_source=CM_QDI_DS_QIAGEN-Discovery-Platform_0526_SEA_Google_Global&gad_source=2&gad_campaignid=21994769191&gclid=Cj0KCQjwh4TVBhCWARIsAG0czmoQDo_QItz6RD8h3LvGp8MfSLJLJvo95V3Sd64tzfrT3ritfw5536MaAhdIEALw_wcB

Background: Can an Existing Drug Treat Other Diseases?

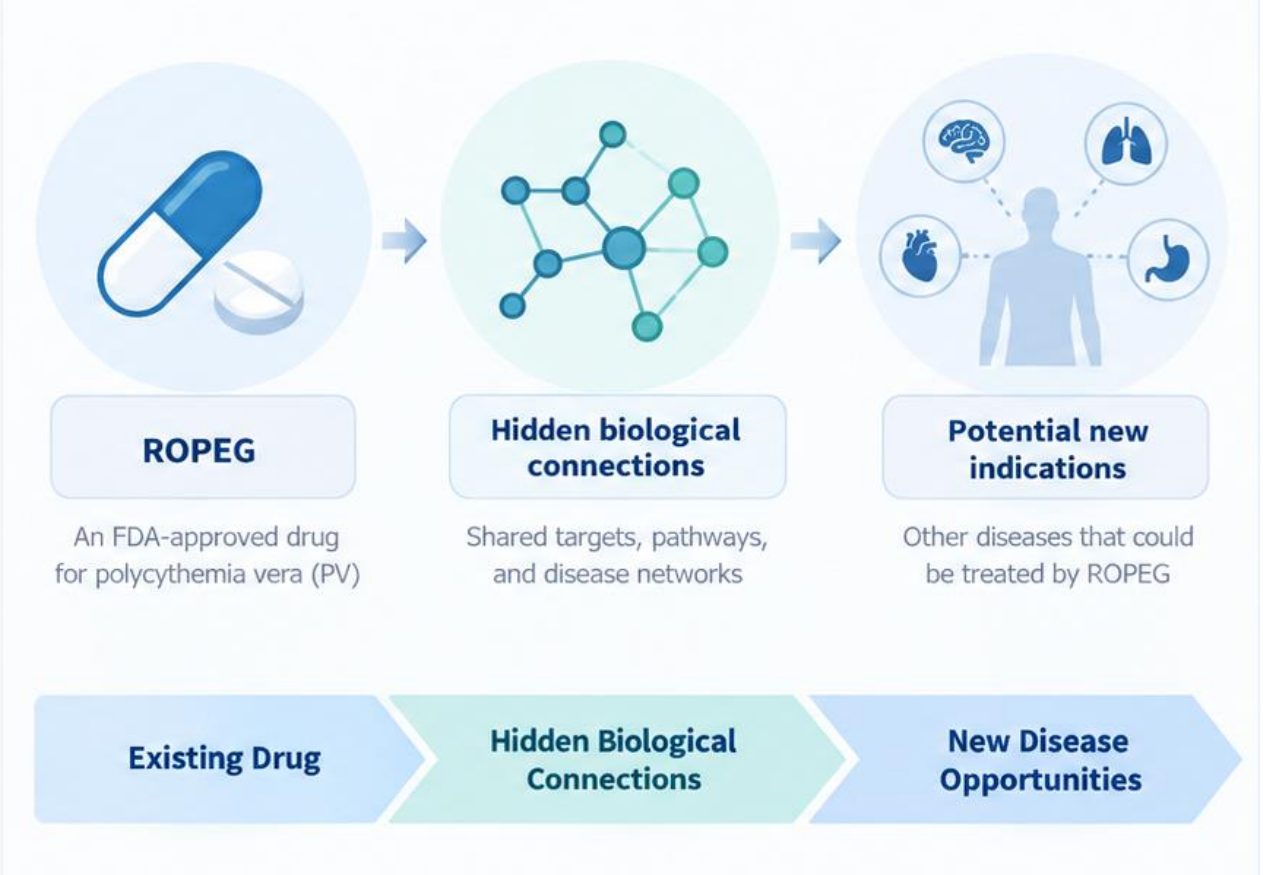
- **ROPEG (Ropeginterferon alfa-2b)** is an FDA-approved drug for treating **polycythemia vera (PV)**, a blood disorder.

➢ **The Question**

Could ROPEG also be effective for other diseases?

➢ **The Task: Drug Repurposing**

Identify **new potential diseases** that could be treated by an **existing approved drug**.

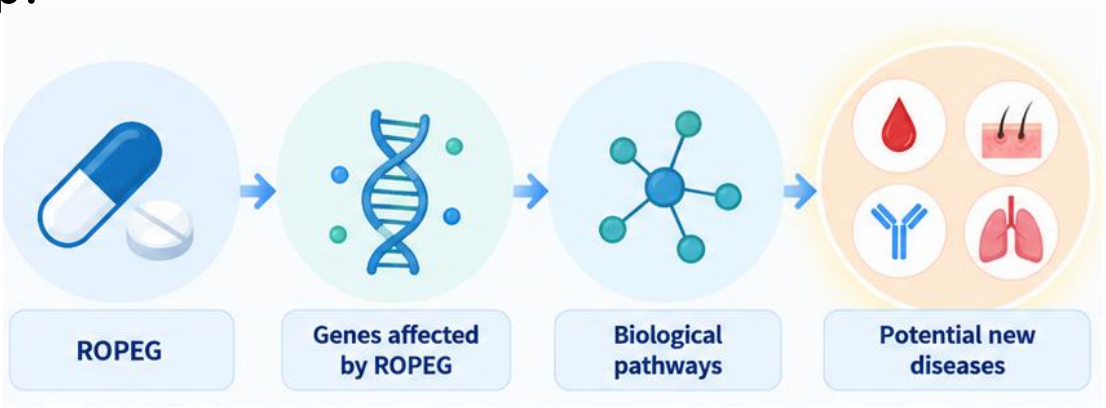


How Does the Knowledge Graph Help?

- Think of a **knowledge graph as a large biomedical map**.
  It connects different types of biomedical entities:

  Drug → Gene → Pathway → Disease

  ROPEG → affects **Gene A**
  → Gene A participates in an **immune-response pathway**
  → this pathway is involved in **Disease X**



- Key Idea

  If **ROPEG affects certain genes**, and those genes are connected to particular pathways and diseases, then :

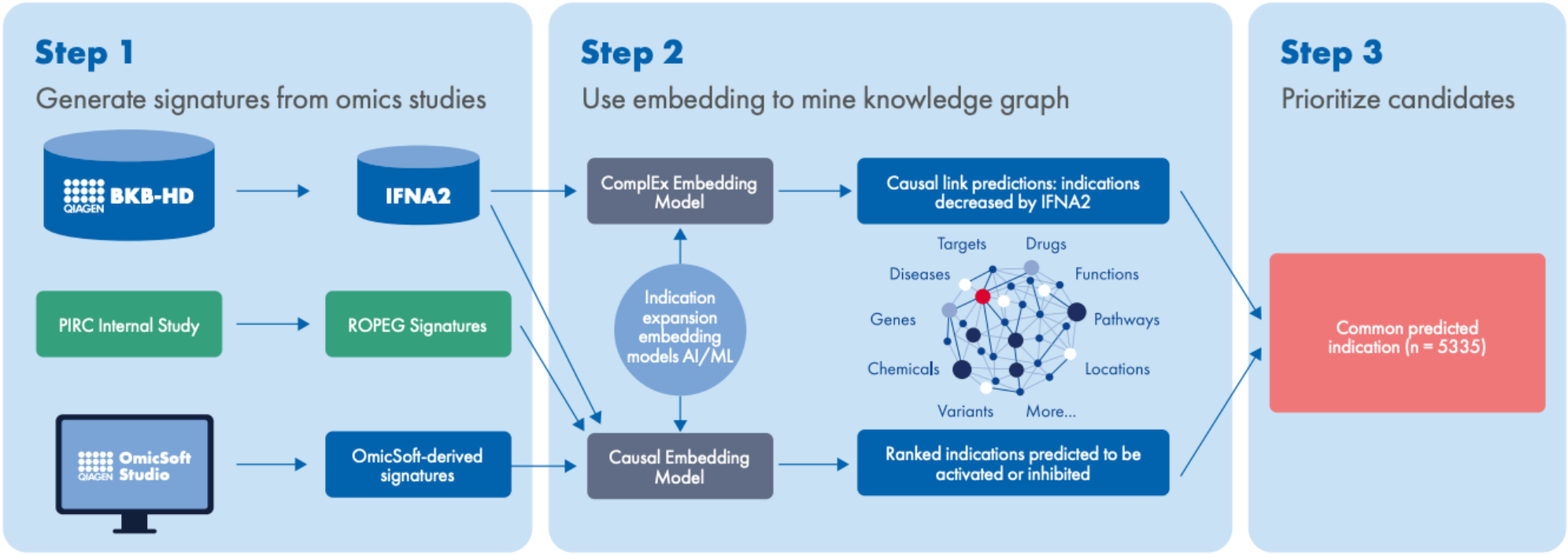
  Which diseases might also be affected by ROPEG?

- Graph Reasoning

  ROPEG → Gene → Pathway → **? Disease**

  Drug A → affects → Gene X → participates in → Pathway Y → related to → Disease B

Workflow



1. Understand the drug
What genes and biological responses does ROPEG affect?

2. Search the knowledge graph
What diseases are connected to these effects?

3. Rank the candidates
Which diseases are most promising for further validation?

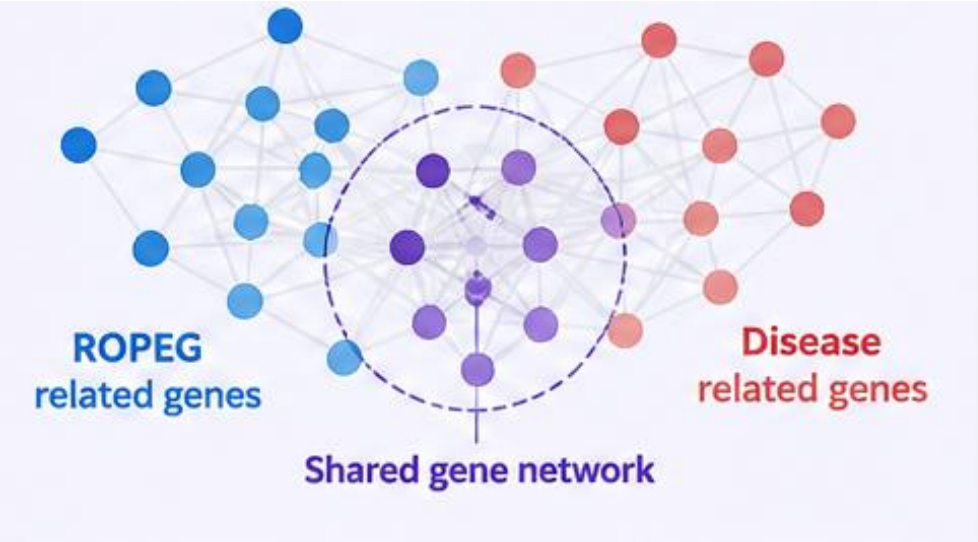
Results: KG Really Helps

➢ **Discovered New Candidate Diseases** 🔍

The knowledge-graph approach highlighted several high-ranking potential indications:

- **Chronic myeloid leukemia**
- **Cutaneous melanoma**
- **Cutaneous T-cell lymphoma (CTCL)**

Importantly, **CTCL showed positive experimental validation**.

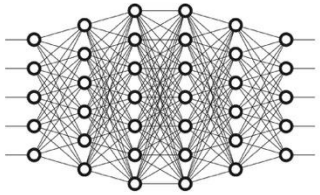


Why Do We Need Knowledge Graphs

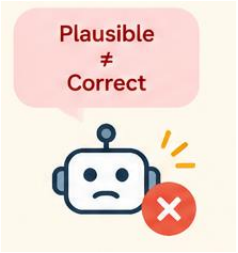
Why Are LLMs Alone Not Enough?

LLMs have several limitations.

- Knowledge is implicit
  Knowledge is encoded in the parameters



- Answer may be unreliable
  LLMs suffer from hallucination



- Reasoning is hard to trace
  Reasoning process is a black box



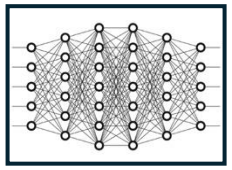
- Knowledge is difficult to update
  Require fine-tuning



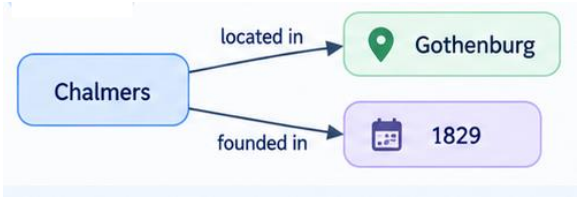
How Knowledge Graphs Address These Limitations

- Explicit & Structured Knowledge

  We know how things are connected.



VS

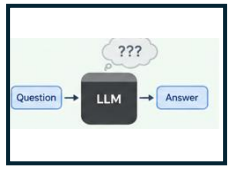


LLM

KG

- Traceable Connections

  The connection path is visible

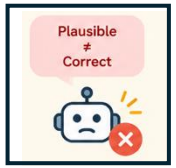


VS

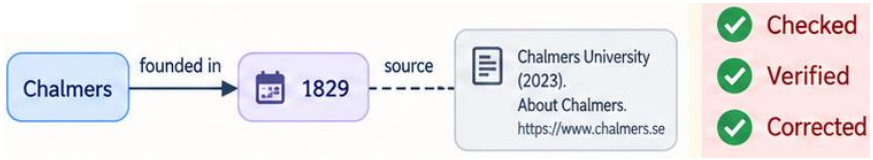


- More Reliable & Verifiable

  Facts in KG can be checked

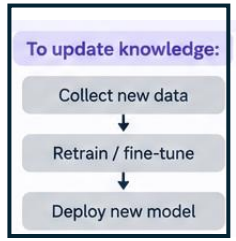


VS

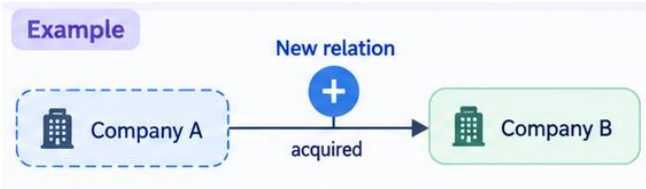


- Easy to Update & Expand

  Insert new nodes and relations directly.



VS



How to Combine Knowledge Graph with LLMs

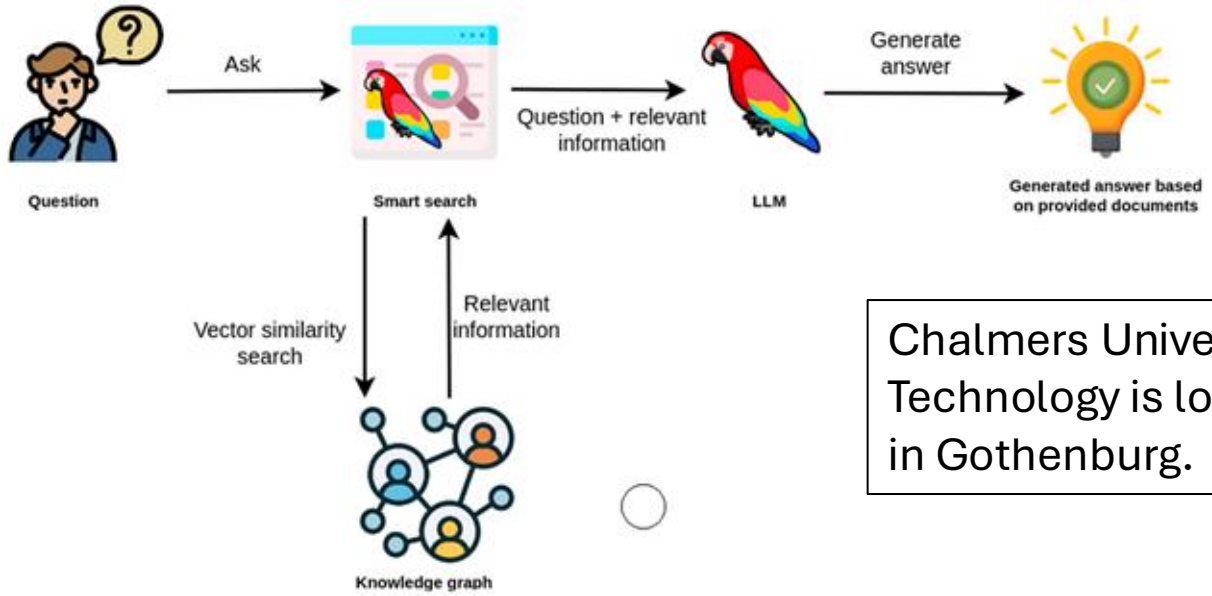
How to Use a Knowledge Graph for Question Answering

➢ Steps

1. Understand the Question
2. Retrieve Relevant Knowledge
3. Add Knowledge to the Prompt
4. Generate the Answer with an LLM

Where is Chalmers located?

Chalmers → locatedIn → Gothenburg



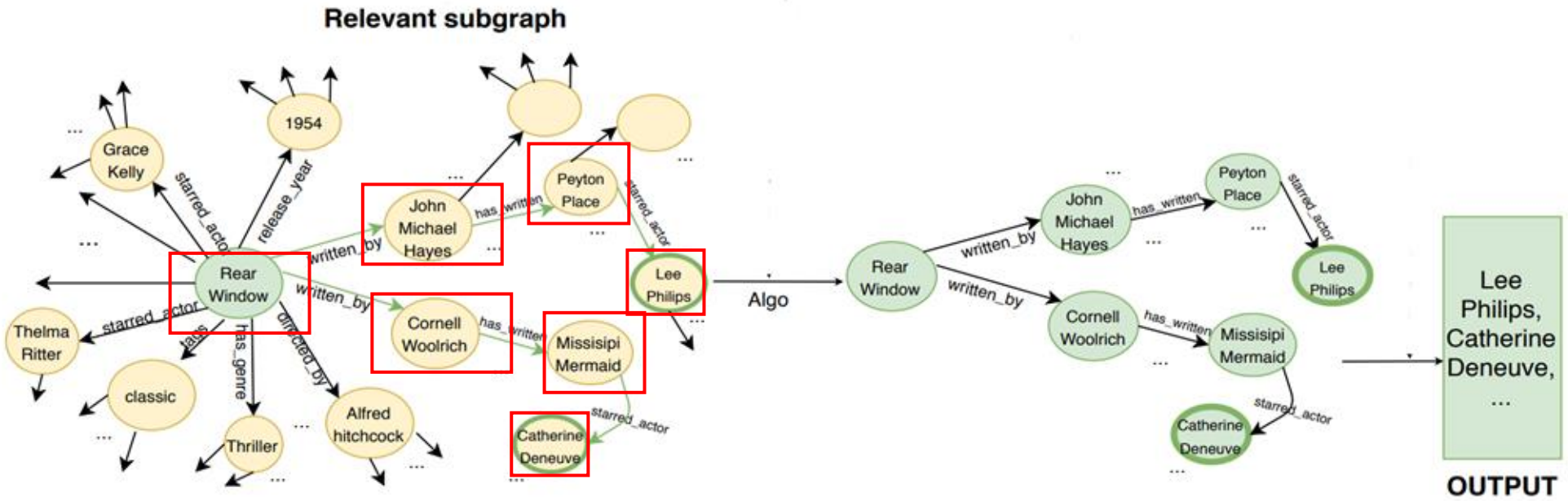
Chalmers University of Technology is located in Gothenburg.

❖ KG finds the knowledge. LLM explains the knowledge.

Multi-hop Question Answering

➢ Question

Who starred films for the screenwriter of Rear Window?



Step 1 — Identify the Topic Entity
Rear Window

Step 2 — Hop 1: Find the Writer

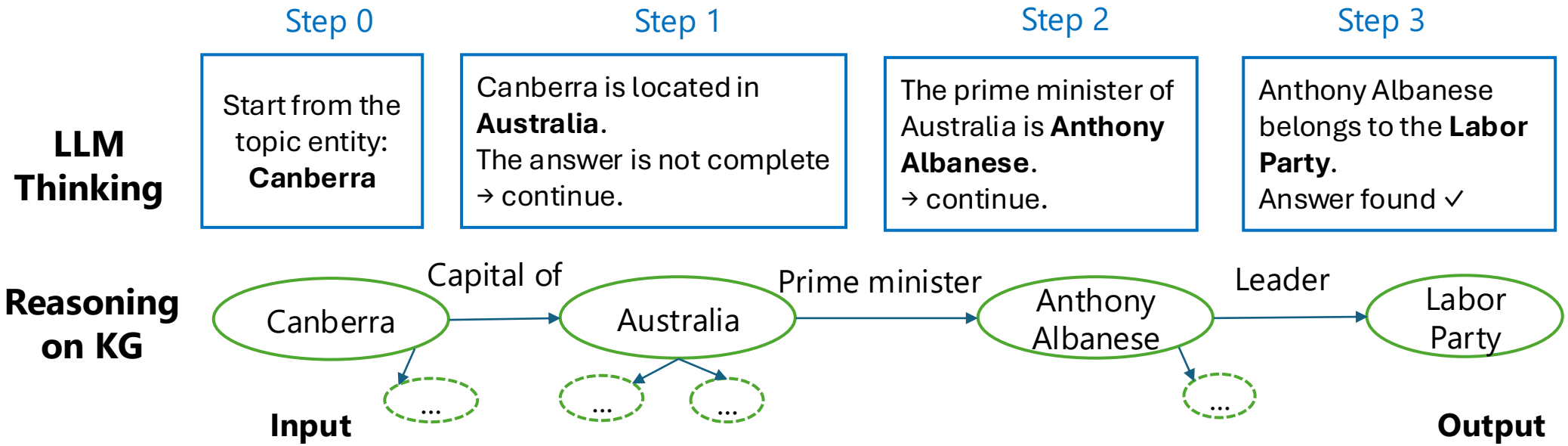
Step 3 — Hop 2: Find Other Films

Step 4 — Hop 3: Find the Actors

Reasoning over Knowledge Graphs

- Question

Which political party does the prime minister of the country where Canberra is located belong to?



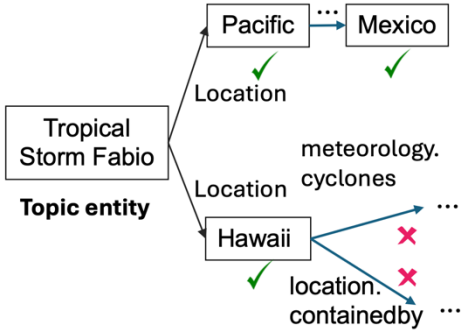
- LLM decides where to go; KG provides explicit facts and connections.

My Work: Iterative Question Guidance

➢ **Motivation**

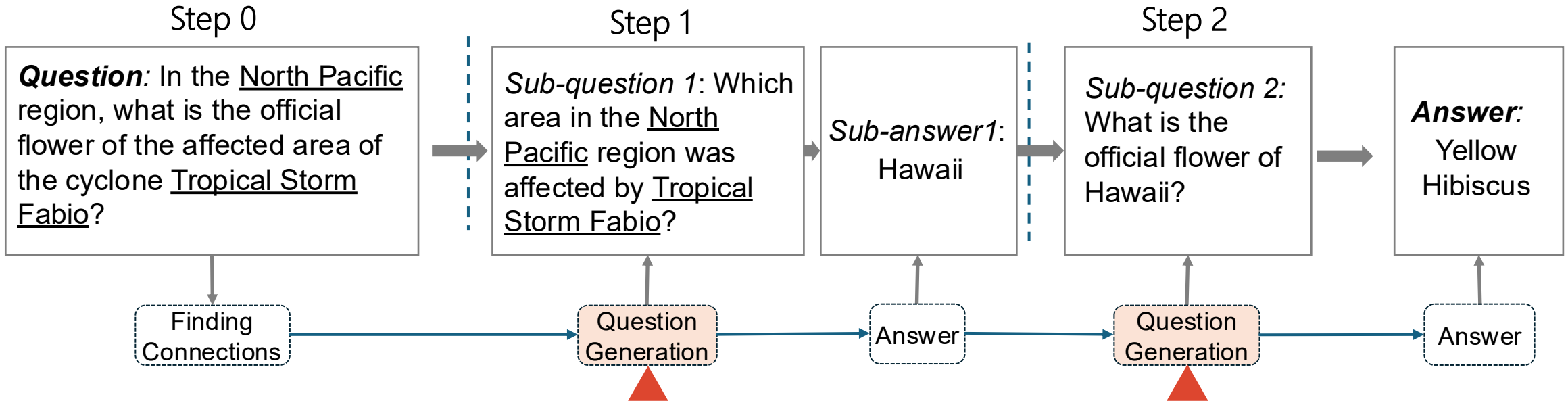
Complex questions contain **multiple entities, and reasoning goals**. LLMs may:

- be distracted by irrelevant information,
- explore too many possible paths



➢ **Key Idea: Ask One Question at a Time**

we iteratively decompose the whole complex question into focused sub-questions.



❖ We use sub-questions to tell it where to look next.

My Work: Using GNNs for Knowledge Graph Reasoning

- **Motivation**

  Looking at only the current node may miss important context.

  Do not judge a node only by who it is, also look at who it is connected to.

- **Two-Hop Neighbor Aggregation with GNN**

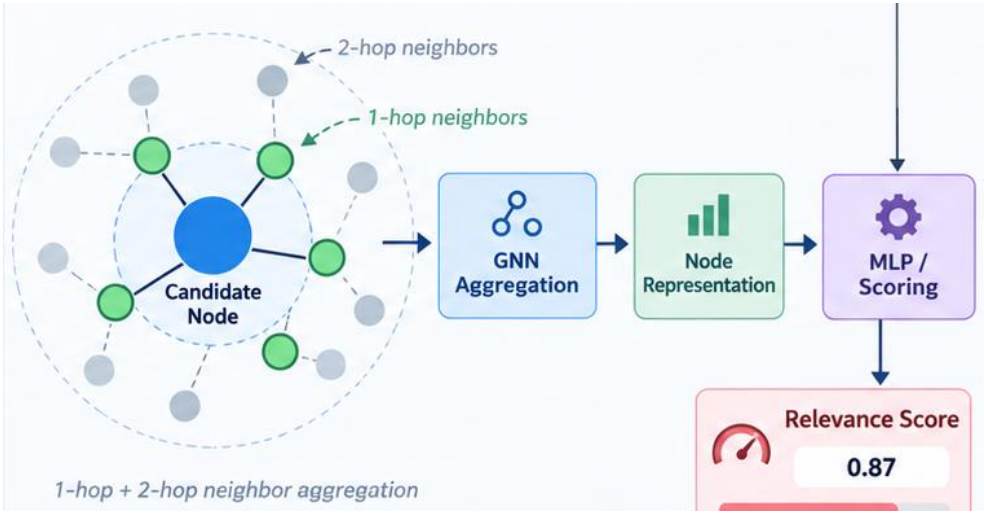
Use GraphSAGE-style aggregation (Hamilton et al., 2017):

$$\hat{\mathbf{h}}_{1h} = \sigma\Big(\mathbf{W}\cdot[\mathbf{h}_{1h}||\mathrm{AGG}\{\mathbf{h}_{2h} \mid e_{2h} \in \mathcal{N}(e_{1h})\}]\Big)$$

Pass the vector through a two-layer MLP to classify: *Relevant* *vs.* *Irrelevant*

$$\hat{\mathbf{y}} = \mathrm{Softmax}\big(\mathbf{W}_2\,\sigma(\mathbf{W}_1\,\mathbf{h} + \mathbf{b}_1) + \mathbf{b}_2\big)$$

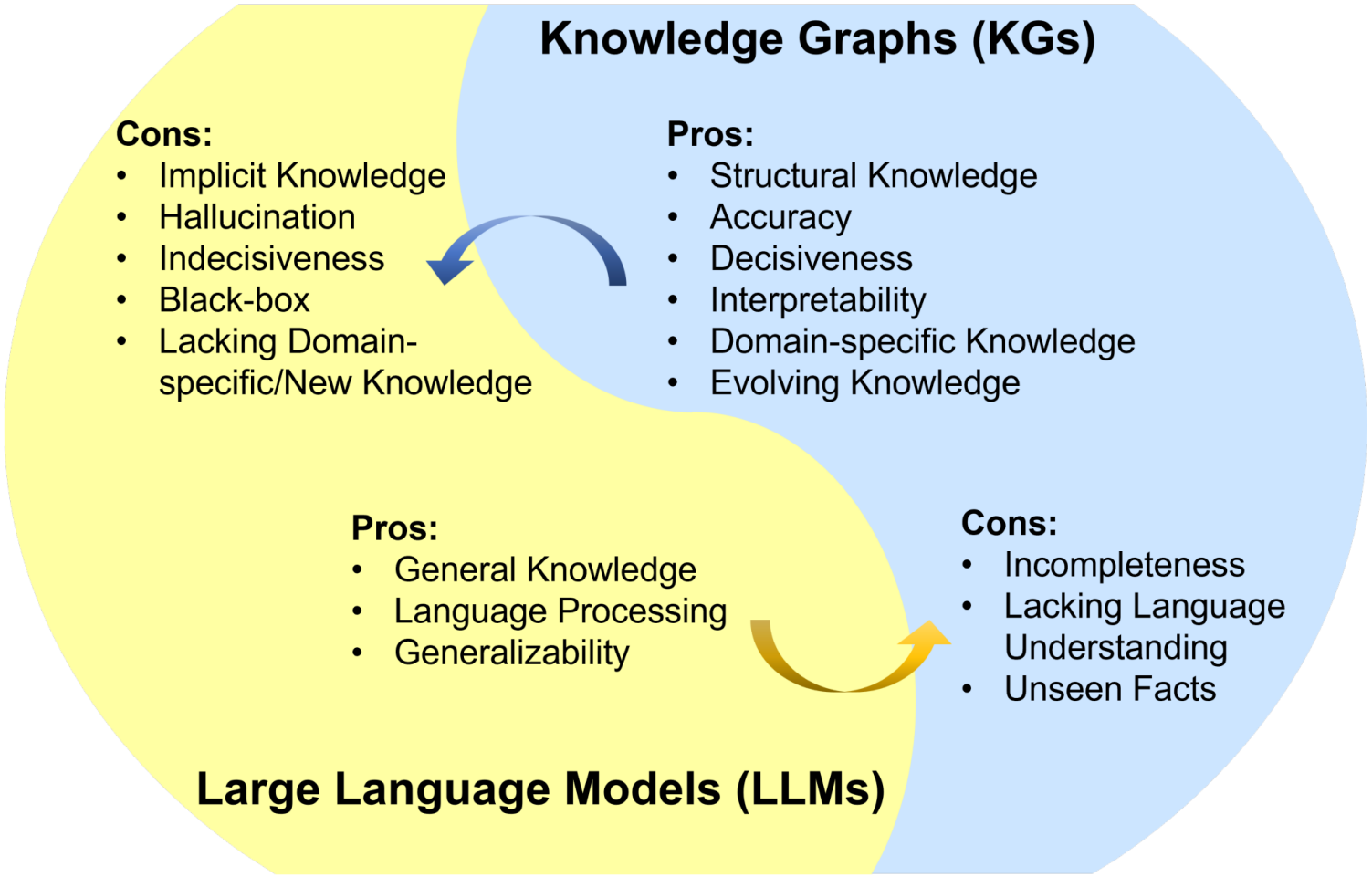
Question



❖ GNNs help us decide where to look in the graph.

LLMs and KGs Have Complementary Strengths

➢ **Unifying LLMs and knowledge graphs**



❖ LLMs provide intelligence to reason; KGs provide the knowledge to reason with.

Thank you!

Happy to take questions.